A mixture cure model for amyloid accumulation

Notation, priors, and first results

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1101 ADNI subjects, 3393 amyloid PET scans

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1 The idea

A standard progression model assumes everyone is on the same journey and differs only in how far along they are. This model makes two weaker assumptions instead.

1. **Some people never accumulate at all.** They sit at a constant level for life. We never observe which people these are, so the model treats it as a latent class.
2. **Among those who do accumulate, they start at different ages.** Before that age they are flat; after it they follow a shared trajectory.

Together these give three states a subject can occupy at any age: *never will accumulate*, *will but has not started*, and *accumulating*. The first two look identical in data — both are flat — which is the central difficulty and the reason the model needs care.

The name comes from survival analysis. A *cure model* is a survival model in which some fraction of the population never experiences the event; the survival curve flattens out at a positive value instead of decaying to zero. Here the "event" is the onset of amyloid accumulation.

2 Notation

Every symbol used in the paper, in one place. Age is always calendar age in years. Amyloid level is always SUVR (figures show Centiloid, a linear rescaling).

Symbol	Range	Meaning
Indices and data		
i	$1, \ldots, 1101$	subject
j	$1, \ldots, n_i$	scan within subject (n_i from 2 to 7)
a_{ij}	years	age of subject i at scan j
Y_{ij}	SUVR	observed amyloid at that scan
$\tilde{a}_i$	years	subject's *visit centroid*, $\tilde{a}_i = \overline{a_{ij}}$
$x_i,\ w_i$	—	covariates (APOE4 carriage, female sex), centred
y^*	0.746 SUVR	positivity threshold (= 24 Centiloid)
Latent per-subject quantities		
Z_i	$\{0, 1\}$	**susceptibility**: 1 if the subject ever accumulates
α_i	years	**departure age**: when accumulation begins. Never observed

Symbol	Range	Meaning
δ_i	real	**log speed**: the subject travels at e^{δ_i}
b_i	SUVR	level of a *non*-accumulator, held for life
$\tilde{x}_i$	SUVR	subject's amyloid level at their own centroid $\tilde{a}_i$
$A_i(a)$	SUVR	the subject's amyloid at age a
Population parameters		
ϕ	$\mathbb{R}^3$	covariate effects on **susceptibility**
π_i	$(0,1)$	$\Pr(Z_i = 1)$, from $\operatorname{logit} \pi_i = w_i^\top \phi$
γ	$\mathbb{R}^3$	covariate effects on **departure age**
σ_α	> 0	spread of departure ages
$\alpha_{\min}, \alpha_{\max}$	25, 125	fixed bounds on departure age
ψ	$\mathbb{R}^2$	covariate effects on **speed**
σ_δ	> 0	between-subject spread of log speed
μ_b	SUVR	the **floor**: both the launch level and the non-accumulator mean
σ_b	> 0	spread of non-accumulator levels
θ	$\mathbb{R}^{10}$	spline coefficients of the **rate curve**
ν	$\mathbb{R}^8$	spline coefficients of the **measurement scale**
$r_A(y)$	SUVR/yr	rate of accumulation at level y
$G_A(y)$	years	the **clock**: disease-years from y to threshold

3 The model, one layer at a time

3.1 Who accumulates

$$Z_i \sim \text{Bernoulli}(\pi_i), \qquad \operatorname{logit} \pi_i = w_i^\top \phi. \tag{1}$$

Z_i is never observed. It is *marginalised* out of the likelihood, which means we sum over both possibilities weighted by their probabilities rather than guessing which one holds.

3.2 Those who do not: a constant level

$$A_i(a) \equiv b_i \quad \text{for all } a, \qquad b_i \sim \mathcal{N}(\mu_b,\ \sigma_b^2) \ \text{ truncated to } b_i < y^*. \tag{2}$$

The truncation matters: a subject seen *above* threshold has certainly accumulated, so the non-accumulator branch must not be able to reach them.

3.3 Those who do: flat, then a shared curve

Accumulation is governed by a single rate curve — how fast amyloid grows depends only on how much is already there:

$$\frac{dA_i}{da} = e^{\delta_i}\, r_A\big(A_i(a)\big), \qquad \log r_A(y) = B_A(y)^\top \theta, \tag{3}$$

where B_A is a cubic B-spline basis. Every subject shares r_A; e^{δ_i} only rescales how fast they move along it.

Before departure the subject is held at the floor, so the full trajectory is

$$A_i(a) = \begin{cases} \mu_b, & a < \alpha_i \quad \text{(the } \textit{clamp}\text{)} \\ G_A^{-1}\left\{G_A(\mu_b) + e^{\delta_i}(a - \alpha_i)\right\}, & a \geq \alpha_i, \end{cases} \tag{4}$$

with the clock

$$G_A(y) = \int_{y^*}^{y} \frac{du}{\max\{r_A(u),\, r_{\min}\}}. \tag{5}$$

Why the clock appears. Because the rate depends only on level and not on age, the ODE can be separated and integrated exactly. (5) is the result: $G_A(y)$ is the number of disease-years needed to travel from y to threshold at unit speed. On this scale the trajectory is a straight line, which is why (4) has a closed form and requires no numerical integration. Two consequences worth remembering:

- $-G_A(\mu_b)$ is the **transit budget**: the time an accumulator must spend getting from the floor to positivity. Fitted here at **11.0** years.
- Accumulators have **no subject-specific intercept**. Everyone launches from the same μ_b. This is deliberate: given a free starting level *and* a departure age, a lower start traded against an earlier departure leaves every fitted value unchanged, so the two would be unidentifiable. Fixing the launch level removes that.

A note on where the initial condition comes from. An ODE needs an initial condition, and the usual choice in progression models is a *level*: give each subject a free starting value, estimated from their own data. This model does not. Every accumulator starts from the same level μ_b, and what varies between subjects is *when* they start. The initial condition is therefore supplied by the prior on the departure age α_i in the next section, not by a per-subject level parameter.

The reason is that the two cannot both be free. Written out, the trajectory depends on a subject-specific starting level b_i and a departure age only through the single combination $G_A(b_i) - e^{\delta_i}\alpha_i$: a lower starting level traded against an earlier departure leaves *every fitted value unchanged*. Estimating both would leave an exact flat direction in the likelihood, closed only by whatever the priors happen to say. Fixing the launch level at μ_b removes it, and the substantive gain is that "this subject is further along than expected" is then explained by a mechanism — they left earlier — rather than by a free parameter.

3.4 When they start

$$\alpha_i = \alpha_{\min} + \exp\left(x_i^\top \gamma\right) E_i^{\sigma_\alpha}, \qquad E_i \sim \text{Exponential}(1), \tag{6}$$

which is a Weibull accelerated failure time (AFT) model with scale $\exp(x_i^\top \gamma)$ and shape $1/\sigma_\alpha$. A covariate that doubles $\exp(x^\top \gamma)$ doubles the time to departure — hence "accelerated failure time". Coefficients are read as **time ratios**, e^{γ_q}.

3.5 How fast they go

$$\delta_i = x_i^\top \psi + \sigma_\delta z_i, \qquad z_i \sim \mathcal{N}(0, 1). \tag{7}$$

There is no intercept. A constant in δ would be indistinguishable from raising the whole rate curve, so omitting it is what fixes the scale of the clock. Because e^{δ_i} multiplies r_A, a subject who

has not departed has no trajectory for ψ to act on — so ψ is automatically an effect *conditional on having started*.

3.6 How the two curves are regularised

Both r_A and ω_A are splines with more coefficients than the data could pin down independently. Neither is estimated freely: each carries a **random-walk prior** on its coefficients, saying that neighbouring coefficients should be similar and therefore that the curve should be smooth. This is the regularisation that makes a 10-coefficient curve estimable from data that only visit part of the range.

The rate curve: an *adaptive* random walk. The increment variance is itself estimated, so the curve can be stiff where data are thin and flexible where they are not. Writing the truncations explicitly, since they are doing real work:

$$\theta_j \sim \mathcal{N}\big(\theta_{j-1},\ \varsigma_j^2\big)\,, \qquad j = 3, \ldots, 10, \tag{8}$$

$$\varsigma_j = \sqrt{\lambda_j \,/\, \varsigma_0}, \tag{9}$$

$$\lambda_j \sim \text{Exponential}(1)\ \mathbb{1}\{0.01 < \lambda_j < 100\}, \tag{10}$$

$$\varsigma_0 \sim \text{Gamma}(1, 1)\ \mathbb{1}\{0.01 < \varsigma_0 < 100\}. \tag{11}$$

The indicator notation is deliberate: λ_j and ς_0 are drawn from *truncated* distributions, restricted to those intervals. Nothing is applied to ς_j afterwards — it is whatever $\sqrt{\lambda_j/\varsigma_0}$ gives, which the truncations confine to $(0.01,\ 100)$.

Read the two variance components as doing different jobs. ς_0 is a *global* smoothing level shared by the whole curve: large ς_0 means small increments everywhere, hence a smooth curve. λ_j is a *local* multiplier attached to one increment, letting the curve bend sharply in one place without forcing it to bend everywhere. Both are estimated, so how much to smooth is learned rather than chosen.

Why the truncation on λ_j matters. It is not a numerical convenience. The increment SD is $\sqrt{\lambda_j/\varsigma_0}$, so as $\lambda_j \to 0$ that SD goes to zero and the normal density evaluated at its own mean goes to *infinity* — the log-density grows like $-\frac{1}{2}\log\lambda_j$, without bound. An Exponential(1) density is finite at zero, so the untruncated prior does nothing to stop this: a sampler could buy unlimited prior density by pinning two adjacent coefficients together, and spend it on an arbitrarily bad fit elsewhere. We have watched exactly that happen. Truncating λ_j away from zero removes the singular point while leaving the adaptivity intact — the increment SD may still vary by orders of magnitude across the curve.

The measurement scale: a *fixed-width* random walk.

$$\nu_1 \sim \mathcal{N}(-3.4,\ 1.0^2)\,\mathbb{1}\{-9 < \nu_1 < 1\}, \qquad \nu_v \sim \mathcal{N}\big(\nu_{v-1},\ 0.5^2\big)\,\mathbb{1}\{-9 < \nu_v < 1\}, \quad v = 2, \ldots, 8. \tag{12}$$

Here the increment width is *fixed* at 0.5 rather than estimated. The measurement scale is a smooth, slowly varying function of level — there is no reason to expect a kink — so the extra flexibility of an adaptive walk is unnecessary, and a fixed width is one fewer thing to identify. The intercept ν_1 is centred at $\log 0.033$, the assay's published repeatability, anchoring the curve at the low end where subjects barely move and the data would otherwise say little.

No further shape constraint. Beyond these random walks, nothing else is imposed on either curve. Their shape is determined by the data.

3.7 What we measure

$$Y_{ij} \sim t_\nu(\mu_{ij},\ \omega_A(\mu_{ij}))\,, \quad \nu = 4, \qquad \log \omega_A(y) = B_A^v(y)^\top \nu, \tag{13}$$

where $\mu_{ij} = A_i(a_{ij})$. Measurement error is indexed by *level*, not by disease stage, because scan noise is a property of how much signal there is. Student-t rather than Gaussian makes the fit robust to occasional bad scans. Note ω is a *scale*, not a standard deviation: the SD is $\omega\sqrt{\nu/(\nu-2)} = 1.414\,\omega$.

3.8 Putting it together

For each subject the two branches are combined on the log scale:

$$\ell_i = \log\Big\{ \underbrace{(1-\pi_i)\textstyle\prod_j p(Y_{ij} \mid b_i)}_{\text{never accumulates}} + \underbrace{\pi_i \textstyle\prod_j p\big(Y_{ij} \mid A_i(a_{ij})\big)}_{\text{accumulates}} \Big\}. \tag{14}$$

This is what "marginalising Z_i" means in practice. It is computed with the log-sum-exp trick, never by exponentiating first, because for a badly fitting subject one branch can be hundreds of log-units below the other.

4 Priors

All covariates are centred, so intercepts refer to a subject at the cohort average.

Parameter	Prior	Reasoning
Susceptibility		
ϕ_1	$\mathcal{N}(0.95,\ 1.0^2)$	$\text{logit}^{-1}(0.95) = 0.72$: a weak belief that most people accumulate
$\phi_q,\ q > 1$	$\mathcal{N}(0,\ 0.5^2)$	one SD is an odds ratio of 1.6
Departure age		
γ_1	$\mathcal{N}(\log 45,\ 0.75^2)$	centres median departure near 70 years
$\gamma_p,\ p > 1$	$\mathcal{N}(0,\ 0.25^2)$	one SD is a time ratio of 1.28
σ_α	$\text{Gamma}(2, 8)\mathbb{1}\{0 < \sigma_\alpha < 2\}$	log-scale spread of onset; bounded away from a funnel
$\alpha_{\min}, \alpha_{\max}$	25, 125 fixed	departure cannot precede childhood or exceed a lifespan
Speed		
ψ_q	$\mathcal{N}(0,\ 0.25^2)$	one SD is a rate ratio of 1.28
σ_δ	$\mathcal{N}^+(0, 1)\mathbb{1}\{0 < \sigma_\delta < 2\}$	half-normal; 2 would be a 7-fold speed range
z_i	$\mathcal{N}(0, 1)\mathbb{1}\{\lvert z_i\rvert < 4\}$	standard normal, bounded for numerical safety
Floor		
μ_b	$\mathcal{N}(0.45,\ 0.15^2)\mathbb{1}\{Y_L < \mu_b < y^*\}$	must lie below threshold
σ_b	$\mathcal{N}^+(0,\ 0.15)\mathbb{1}\{0 < \sigma_b < 0.5\}$	spread of flat levels
Curves		
θ_1	fixed at -6.9	its basis support lies entirely below μ_b, where nothing is evaluated

Parameter	Prior	Reasoning
θ_2	$\mathcal{N}(-6.9,\ 2.0^2)$	starts the rate-curve random walk
$\theta_j,\ j \geq 3$	adaptive RW	**see §3.6** — increments $\varsigma_j = \sqrt{\lambda_j/\varsigma_0}$
λ_j	$\text{Exp}(1)\mathbb{1}\{0.01 < \lambda_j < 100\}$	*local* smoothing multiplier; truncated away from 0 to remove a singularity
ς_0	$\text{Gamma}(1,1)\mathbb{1}\{0.01 < \varsigma_0 < 100\}$	*global* smoothing level
ν_1	$\mathcal{N}(-3.4,\ 1.0^2)\mathbb{1}\{-9 < \nu_1 < 1\}$	$e^{-3.4} = 0.033$, the assay repeatability floor
$\nu_v,\ v \geq 2$	$\mathcal{N}(\nu_{v-1},\ 0.5^2)\mathbb{1}\{-9 < \nu_v < 1\}$	*fixed*-width RW; no adaptivity needed

5 Results

Three chains of 10,000 iterations. Largest $\hat{R}$ is 1.073, median 1.006.

5.1 The rate curve, and survival with and without the cure component

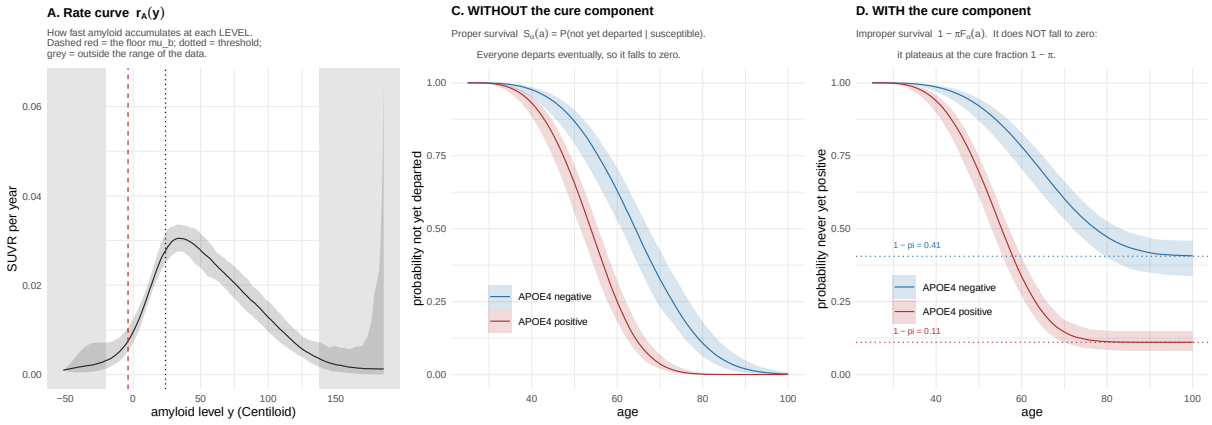


Figure 1: (A) The rate curve $r_A(y)$. (B) Survival among susceptibles only — a proper survival function. (C) Survival in the population, which plateaus at the cure fraction.

Panel A is the rate curve. Accumulation is slow at low burden, fastest around 36 Centiloid, and slows again above that — amyloid saturates. By 137 CL the rate has fallen to 13% of its peak. The grey regions are outside the range of the observed data, where the curve is held up by the random-walk prior of §3.6 rather than by observations.

Panels B and C make the distinction the model exists for.

This is the distinction the model exists to make.

Panel B shows $S_\alpha(a) = \Pr(\alpha_i > a \mid Z_i = 1)$: among people who *will* accumulate, the fraction who have not yet started. It is a proper survival function and falls to zero — everyone in this group eventually departs.

Panel C shows the population version, $1 - \pi\, F_\alpha(a)$: the fraction who have not started, *including* those who never will. It does not reach zero. It flattens at $1 - \pi$, the **cure fraction**:

	cure fraction $1-\pi$	median age at crossing y^*
APOE4 negative	0.404	77.0 [75.0, 79.2]
APOE4 positive	0.112	64.9 [63.8, 66.3]

So 40% of non-carriers never accumulate, against 11% of carriers; and among those who do, carriers reach positivity **12.1** years earlier.

A plateau is what distinguishes a cure model from an ordinary survival model. Fitting these data with a standard AFT — panel B's assumption applied to everyone — would force the curve to zero and, to do so, would have to push the remaining departures to implausibly late ages.

5.2 Covariate effects: three separate channels

Because the model has three places a covariate can act, it can say *how* a risk factor works rather than only that it matters.

Channel	Covariate	Estimate [95% CrI]	Reads as
Susceptibility (ϕ)	APOE4	OR **5.44** [3.72, 8.12]	odds of ever accumulating
	female	OR 1.48 [1.03, 2.11]	
Departure (γ)	APOE4	TR **0.729** [0.679, 0.778]	27% earlier onset
	female	TR 0.955 [0.889, 1.020]	
Speed (ψ)	APOE4	RR 1.097 [0.972, 1.242]	*includes 1*
	female	RR 0.906 [0.806, 1.005]	

APOE4 acts on whether a subject accumulates, and on when they start — not on how fast they go once started. The rate effect covers one. This is the kind of decomposition a single-channel model cannot make: with only a level or a rate parameter available, "more carriers accumulate" and "carriers start sooner" both have to be expressed as "carriers are further along", and the distinction is lost.

5.3 Aligning subjects to their own onset

Left: calendar years since departure. Everyone starts at the floor at $t = 0$, then fans apart, because each travels at their own e^{δ_i}. The three black curves are the population trajectory at -1, 0 and $+1$ SD of δ.

Right: the same data on the *disease clock*, $e^{\delta_i}(a - \alpha_i)$. The model says this is the coordinate on which all between-subject differences vanish, and the fan collapses onto one curve. Residual spread about it is 0.059 SUVR.

Read together, the two panels are the model's central claim: **one shared trajectory, entered at different ages and traversed at different speeds.** The median departure age is 57 years — decades before most subjects were first scanned, which is why so little data lies to the left of $t = 0$.

6 Next: extending to tau

The same machinery extends to a second biomarker, with one addition: tau's rate is modulated by *concurrent amyloid*.

$$\frac{dT_i}{da} = \exp\Big\{\delta_i^T + \underbrace{s\big(A_i(a)\big) - s(y_A^*)}_{\text{amyloid drives tau}}\Big\}\, r_T\big(T_i(a)\big), \qquad s(y) = B_A(y)^\top \varphi. \tag{15}$$

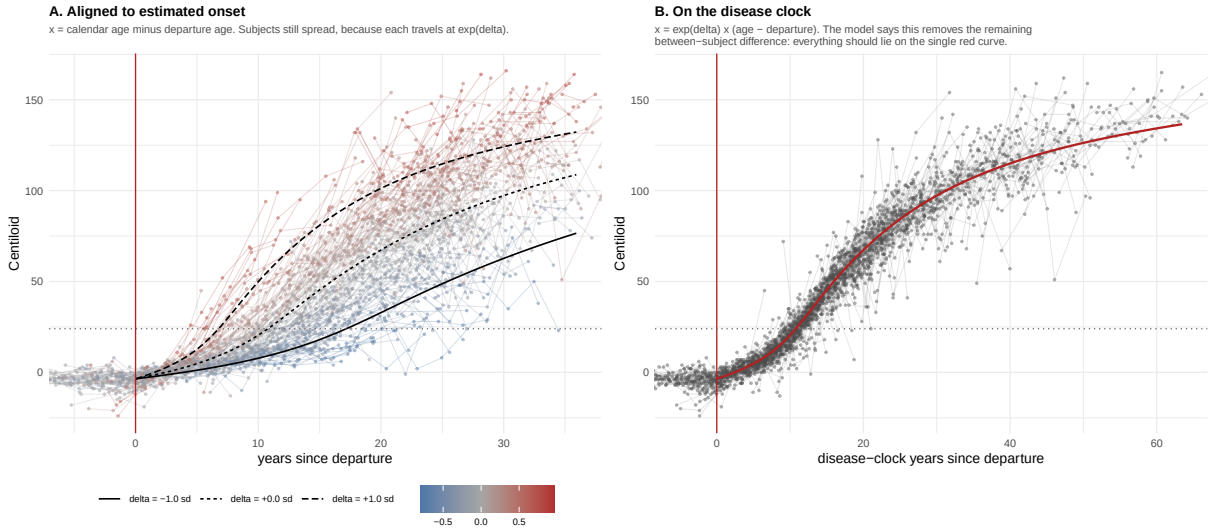


Figure 2: All 785 subjects with posterior probability of susceptibility above 0.5, aligned to their own estimated departure age.

Tau then advances on a clock warped by the amyloid path, $\Psi_i(a) = \int \exp\{s(A_i(u))\}\, du$, and $s(\cdot)$ is the scientific target: it states what lowering amyloid would do to tau velocity. The normalisation $s(y_A^*) = 0$ is a convention fixing the scale, not a finding.

Three things to settle before fitting it.

- **One susceptibility indicator or two?** Two would allow an amyloid non-accumulator with accumulating tau, which contradicts the cascade the coupling encodes. One forces $\pi_A = \pi_T$, which the data contradict: the observed lower bounds are $\pi_A \geq 0.51$ and $\pi_T \geq 0.33$.
- **Tau data are thinner.** 444 subjects with tau scans, of whom only 44 are observed to cross threshold. Covariate effects on tau susceptibility will be largely prior-driven.
- **Cost.** The coupling integral and three further curve blocks roughly double the time per iteration.